MATH 213 – Homework 9

Before beginning, please (and this is **Very Important!!!)**

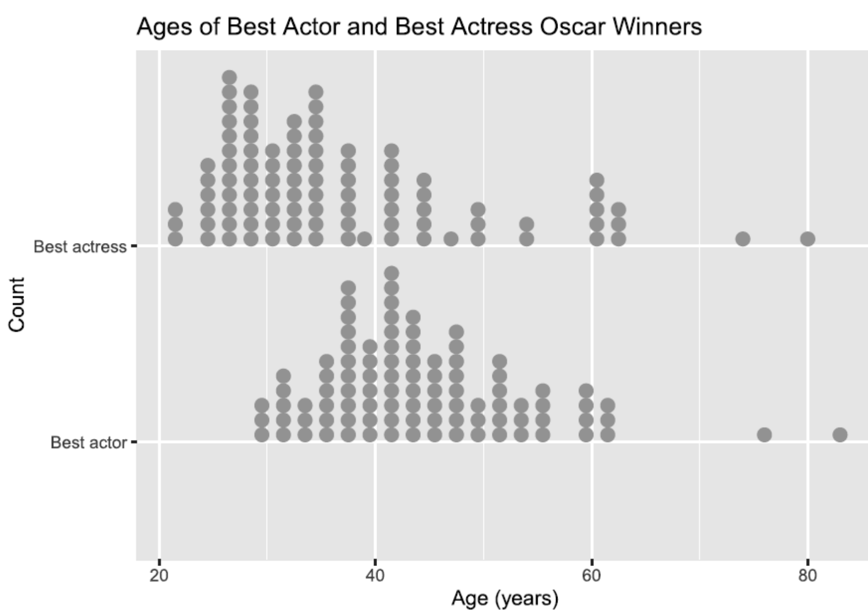
- Read the feedback I have left for you on Homework 6 in Blackboard.
- Read the solutions for Homework 6 in the Solutions section of Blackboard.
- Consult the guidance for writing about distributions of numerical variables from class

The following graph and summary statistics are calculated from the Oscars dataset we have already considered in class. Write a paragraph or two to compare the distributions of the ages of Best actor and Best actress winners. Compare the groups in terms of our usual key features of such distributions, including shape, center, spread, and outliers.

Start with a who/what/when/where sentence. There are two ways to proceed:

- Either describe both distributions the way we've been practicing, and then compare/contrast the two and say the main features you want to say are similar and/or different.
- Or jump straight in to the main similarity or difference between the ages of the groups you want to highlight and support it with some statistics that are shown below. Then, select the NEXT MOST IMPORTANT THING you want to highlight about the graphs, again supporting it with numbers, and so on.

Here is the graph:



And here are the summary statistics:

```
## # A tibble: 2 x 8
##   award        mean_age median_age sd_age  q025   q25   q75  q975
##   <chr>           <dbl>      <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Best actor       44.4         43   9.61  30.4    38  49.5    62
## 2 Best actress     36.8         33  12.1   22.7    28  41      63
```

The graph shows the ages of academy award winners for Best Actor and Best Actress between 1929 and 2022, highlighting the differences in their age distributions. The typical Best Actor winner is around 45 years old, with ages ranging from the early 20s to over 80 years old. Most winners fall within the 30 to 55-year range, but the distribution is slightly right-skewed, meaning that some winners are significantly older than the majority. This is further shown by the presence of two outliers who are much older than the rest of the Best Actor winners.

Conversely, the distribution for Best Actress winners shows a younger average age of just over 36 years old, with ages ranging from the early 20s to 80 years old. The distribution is more right-skewed compared to the Best Actor distribution, meaning there is greater variation in winners' ages. For example, most Best Actress winners fall between 28 and 41 years old (as shown by the q25 and q75 values), indicating that younger actresses are more commonly awarded. However, the presence of a few much older winners increases the spread of the distribution, making it more skewed.

The summary statistics further support these differences. The interquartile range for Best Actress winners is larger than that of Best Actors, suggesting a wider variation of ages within the middle 50% of winners. Additionally, the q975 value for both categories are similar (around 62-63 years old), showing that while older winners do exist in both groups, the typical Best Actress winner tends to be younger.